

IVOR Paine Memorial Hospital

Database Table Description

CL2005 - Database Systems Lab Project | Phase 2

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This document describes all 25 tables in the IvorPaineHospital database, including column names, data types, key constraints, nullability, and purpose.

Ward

Represents wards in the hospital. Each ward has a unique name and caters to one specialty.

Column Name	Data Type	Key	Nullable	Description
WardName	VARCHAR(20)	PK	NOT NULL	Unique ward identifier
Location	VARCHAR(20)		NULL OK	Physical location in hospital
TotalBeds	INT		NULL OK	Total bed capacity of ward
SpecialtyName	VARCHAR(20)	FK	NOT NULL	References Specialty(SpecialtyName)

Specialty

Stores medical specialties available in the hospital. A ward covers one specialty; a consultant has one specialty.

Column Name	Data Type	Key	Nullable	Description
SpecialtyName	VARCHAR(20)	PK	NOT NULL	Unique name of the medical specialty
Description	VARCHAR(50)		NULL OK	Brief description of the specialty

Nurse

Base entity for all nurses. Subtypes: NightSister, DaySister, StaffNurse, NonRegisteredNurse.

Column Name	Data Type	Key	Nullable	Description
NurseID	INT	PK	NOT NULL	Unique nurse identifier
NurseName	VARCHAR(20)		NULL OK	Full name of nurse
DateOfBirth	DATE		NULL OK	Nurse date of birth
Address	VARCHAR(50)		NULL OK	Residential address
PositionCode	INT	FK	NULL OK	References Position(PositionCode)

NightSister

Subtype of Nurse. Night Sister manages a ward during night shift.

Column Name	Data Type	Key	Nullable	Description
NurseID	INT	PK,FK	NOT NULL	References Nurse(NurseID)
WardName	VARCHAR(20)	FK	NOT NULL	Ward managed; References Ward(WardName)
IsSedationAllowed	BIT		NULL OK	Whether sedation permitted on her shift
ShiftStart	TIME		NULL OK	Time shift begins

DaySister

Subtype of Nurse. Day Sister manages a ward during day shift.

Column Name	Data Type	Key	Nullable	Description
NurseID	INT	PK,FK	NOT NULL	References Nurse(NurseID)
WardName	VARCHAR(20)	FK	NOT NULL	Ward managed; References Ward(WardName)
ShiftStart	TIME		NULL OK	Time day shift begins

DaySisterRound

Multi-valued attribute: ward round times for a Day Sister.

Column Name	Data Type	Key	Nullable	Description
NurseID	INT	PK,FK	NOT NULL	References DaySister(NurseID)
WardRoundTime	TIME	PK	NOT NULL	Scheduled round time

NursePhone

Multi-valued attribute table storing phone numbers for nurses (one nurse can have many numbers).

Column Name	Data Type	Key	Nullable	Description
NurseID	INT	PK,FK	NOT NULL	References Nurse(NurseID)
PhoneNumber	VARCHAR(20)	PK	NOT NULL	Phone number of nurse

NurseQualifications

Multi-valued attribute table for nurse qualifications (e.g., BSc Nursing, ICU Cert).

Column Name	Data Type	Key	Nullable	Description
NurseID	INT	PK,FK	NOT NULL	References Nurse(NurseID)
Qualification	VARCHAR(50)	PK	NOT NULL	Name of qualification held

Position

Lookup table for staff positions: Student, Junior Houseman, Senior Houseman, AR, Registrar, Consultant.

Column Name	Data Type	Key	Nullable	Description
PositionCode	INT	PK	NOT NULL	Unique position identifier
Description	VARCHAR(50)		NULL OK	Full description of the position
PositionTitle	VARCHAR(20)		NULL OK	Short title/abbreviation

CareUnit

Weak entity (depends on Ward). A ward is subdivided into care units, each with a staff nurse in charge.

Column Name	Data Type	Key	Nullable	Description
CareUnitNo	INT	PK	NOT NULL	Partial key — unique within a ward
CareUnitName	VARCHAR(20)		NULL OK	Descriptive name of care unit
Capacity	INT		NULL OK	Max patients in this unit
WardName	VARCHAR(20)	PK,FK	NOT NULL	References Ward(WardName)
NurseInChargeID	INT	FK	NULL OK	References StaffNurse(NurseID)

StaffNurse

Subtype of Nurse. Assigned to one care unit; in charge of that unit.

Column Name	Data Type	Key	Nullable	Description
NurseID	INT	PK,FK	NOT NULL	References Nurse(NurseID)
CareUnitNo	INT	FK	NOT NULL	References CareUnit
WardName	VARCHAR(20)	FK	NOT NULL	References CareUnit(WardName)
SupervisorRating	INT		NULL OK	Performance rating 1–5

NonRegisteredNurse

Subtype of Nurse. Non-registered nurses assigned to care units; may be in training.

Column Name	Data Type	Key	Nullable	Description
NurseID	INT	PK,FK	NOT NULL	References Nurse(NurseID)
CareUnitNo	INT	FK	NOT NULL	References CareUnit
WardName	VARCHAR(20)	FK	NOT NULL	References CareUnit(WardName)
TrainingStatus	BIT		NULL OK	1 = currently in training

Bed

Each bed has a hospital-wide unique number. Beds belong to a ward.

Column Name	Data Type	Key	Nullable	Description
BedNo	INT	PK	NOT NULL	Unique hospital-wide bed number
BedType	VARCHAR(20)		NULL OK	Type: Standard, ICU, Pediatric, etc.
IsOccupied	BIT		NULL OK	1 = currently occupied
WardName	VARCHAR(20)	FK	NOT NULL	References Ward(WardName)

Doctor

Base entity for all doctors. Consultants are a subtype.

Column Name	Data Type	Key	Nullable	Description
DoctorNo	INT	PK	NOT NULL	Unique doctor identifier
DoctorName	VARCHAR(20)		NULL OK	Full name of doctor
Address	VARCHAR(50)		NULL OK	Residential address
DateOfBirth	DATE		NULL OK	Doctor date of birth

DoctorPhone

Multi-valued: phone numbers for doctors.

Column Name	Data Type	Key	Nullable	Description
DoctorNo	INT	PK,FK	NOT NULL	References Doctor(DoctorNo)
PhoneNumber	VARCHAR(20)	PK	NOT NULL	Doctor phone number

DoctorQualifications

Multi-valued: qualifications held by doctors (e.g., MBBS, FCPS).

Column Name	Data Type	Key	Nullable	Description
DoctorNo	INT	PK,FK	NOT NULL	References Doctor(DoctorNo)
Qualification	VARCHAR(50)	PK	NOT NULL	Qualification title

Consultant

Subtype of Doctor. Consultants hold a specialty and lead a team of doctors.

Column Name	Data Type	Key	Nullable	Description
DoctorNo	INT	PK,FK	NOT NULL	References Doctor(DoctorNo)
SpecialtyName	VARCHAR(20)	FK	NOT NULL	References Specialty(SpecialtyName)

DoctorTeamRecord

Tracks which doctors belong to which consultant's team, their position and tenure.

Column Name	Data Type	Key	Nullable	Description
DoctorNo	INT	PK,FK	NOT NULL	References Doctor(DoctorNo)
DateJoined	DATE	PK	NOT NULL	Date doctor joined this team
EndDate	DATE		NULL OK	Date left team (NULL = current member)
PositionCode	INT	FK	NOT NULL	References Position(PositionCode)
ConsultantNo	INT	FK	NOT NULL	References Consultant(DoctorNo)

PreviousExperience

Records prior work history of doctors at other establishments.

Column Name	Data Type	Key	Nullable	Description
DoctorNo	INT	PK,FK	NOT NULL	References Doctor(DoctorNo)
FromDate	DATE	PK	NOT NULL	Start date at establishment
ToDate	DATE		NULL OK	End date (NULL if ongoing)
Establishment	VARCHAR(20)		NULL OK	Name of hospital/institution
PositionCode	INT	PK,FK	NOT NULL	References Position(PositionCode)

PerformanceHistory

Six-monthly performance grades assigned by the consultant to each team doctor.

Column Name	Data Type	Key	Nullable	Description
DoctorNo	INT	PK,FK	NOT NULL	References Doctor(DoctorNo)
GradeDate	DATE	PK	NOT NULL	Date grade was assigned
PerformanceGrade	VARCHAR(20)		NOT NULL	Grade: Outstanding/Excellent/Good/Satisfactory

PatientPhoneNumber

Multi-valued: phone numbers for patients.

Column Name	Data Type	Key	Nullable	Description
PatientNo	INT	PK,FK	NOT NULL	References Patient(PatientNo)
PhoneNumber	VARCHAR(20)	PK	NOT NULL	Patient contact number

Patient

Core patient entity. Each patient belongs to one ward, one care unit, occupies one bed, and is under one consultant and one doctor in charge.

Column Name	Data Type	Key	Nullable	Description
PatientNo	INT	PK	NOT NULL	Unique patient identifier
PatientName	VARCHAR(20)		NULL OK	Full name
DateOfBirth	DATE		NULL OK	Date of birth
Address	VARCHAR(50)		NULL OK	Residential address
Age	Computed		NULL OK	Derived from DateOfBirth
CareUnitNo	INT	FK	NOT NULL	References CareUnit
WardName	VARCHAR(20)	FK	NOT NULL	References CareUnit(WardName)
DischargeDate	DATE		NULL OK	NULL if still admitted
DateAdmitted	DATE		NULL OK	Date of admission
BedNo	INT	FK	NOT NULL	References Bed(BedNo)
ConsultantNo	INT	FK	NOT NULL	References Consultant(DoctorNo)
DoctorInChargeNo	INT	FK	NOT NULL	References Doctor(DoctorNo)

Complaint

Medical complaints/diagnoses. A patient can have multiple complaints simultaneously.

Column Name	Data Type	Key	Nullable	Description
ComplaintCode	INT	PK	NOT NULL	Unique complaint identifier
Description	VARCHAR(100)		NULL OK	Description of the complaint
Severity	VARCHAR(10)		NULL OK	Low / Medium / High / Critical

Treatment

Catalogue of available treatments in the hospital.

Column Name	Data Type	Key	Nullable	Description
TreatmentCode	INT	PK	NOT NULL	Unique treatment identifier
Description	VARCHAR(100)		NULL OK	Description of treatment
TreatmentType	VARCHAR(30)		NULL OK	Surgical / Medication / etc.
Duration	VARCHAR(20)		NULL OK	Expected treatment duration

TreatmentRecord

Weak entity recording which treatment was given to which patient for which complaint on which date, by which doctor. Enforces: at any time, one treatment per complaint per patient.

Column Name	Data Type	Key	Nullable	Description
PatientNo	INT	PK,FK	NOT NULL	References Patient(PatientNo)
ComplaintCode	INT	PK,FK	NOT NULL	References Complaint(ComplaintCode)
TreatmentCode	INT	PK,FK	NOT NULL	References Treatment(TreatmentCode)
DateStarted	DATE	PK	NOT NULL	Treatment start date
DateEnded	DATE		NULL OK	Treatment end date (NULL if ongoing)
Notes	VARCHAR(500)		NULL OK	Clinical notes
DoctorNo	INT	FK	NOT NULL	Doctor administering treatment